Week 8 — Text Inside-Out

Name: ____ **Date:** ____

> *The 1st letter is number **1**. Slices include **both** ends. And every character is secretly a **number**.*

Today's words

Word	What it means
index	a letter's position number — the 1st letter is number 1
`end`	means "the last position" — `word(end)` is the last letter
`length()`	counts the characters in a string
slice	a cut-out piece, like `word(2:4)` — BOTH end numbers are included
`double()` / `char()`	`double('A')` → 65 reveals the secret number; `char(66)` → B goes backwards

Mini ASCII table (you'll need this!)

A	B	C	D	E	F	G	H	I	J	K	L	M
65	66	67	68	69	70	71	72	73	74	75	76	77

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
78	79	80	81	82	83	84	85	86	87	88	89	90

(space = 32)

1 · Address every letter 📍

Write the index under each letter of MATLAB, then answer:

`M___ A___ T___ L___ A___ B___`

- `word(3)` = __ · `word(end)` = _ · `length(word)` = · `word(2:4)` = ___

2 · X-ray the alphabet 🔍

Use the table (no computer needed — YOU are the computer):

- `double('C')` = __ · `char(72)` = _ · `char(double('M') + 1)` = ___
- Lowercase letters sit exactly 32 higher. So `double('c')` = ______

3 · Crack the number message 🕵️

Each number is one character (remember: 32 = space). Decode:

72 · 73 · 32 · 90 · 79 · 69 → `___ ___ ___ ___ ___ ___`

🧠 Brain teaser (optional — take it home)

A secret message, in pure numbers. Decode it with your table:

89 79 85 · 32 · 70 79 85 78 68 · 32 · 84 72 69 · 32 · 83 69 67 82 69 84

Then write **your own name** as secret numbers and test it on someone at home. Bring both next week for a shout-out — next week we build a 2,000-year-old cipher on top of exactly this trick.